

LUONG PHAT NGUYEN

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 lphatnguyen



EXPERIENCES

AI Research Scientist

Rizlum AI

 Today

 Paris, France

- Research and development of multimodal-model systems for document analysis.
- Contributing to construction of AI training platform and Agent platform.
- Conversational AI with realtime Speech-to-text and Text-to-speech models achieving response of sub-1 second latency.

Postdoctoral Researcher in LLM

Toyota Belgium and LISSI

 July 2024 - March 2025

 Paris, France

- Multimodal dataset analysis (VQA, Conceptual Caption).
- Research and design of multimodal LLM.
- Design of an intelligent emotion agent.

AI Research Scientist

Trax Retail

 Dec 2021 – April 2024

 Paris, France

- Construct Deep Learning models for image recognition (CNN, PyTorch, ResNet).
- Research and development of real-time algorithms in Swift for an iOS mobile application SDK (CoreML, ARKit, SceneKit).
- Research and development of optimization and generalization methods for Deep Learning models (ASAM, SAM, Pruning).

PhD in Computer Vision

Laboratoire d'Informatique Fondamentale et Appliquée de Tours

 Nov 2018 – Dec 2021

 Tours, France

Supervisors: Julien Mille, Nicolas Ragot, Dominique Li and Donatello Conte

Video analysis and classification research with:

- Convolutional and recurrent neural networks (R(2+1)D, R3D, LSTM).
- Self-attention models (Vision/Space-time Transformers).

PUBLICATIONS

Conference Proceedings

- Nguyen, L. P., Abdelkawy, H., & Othmani, A. (2025). Emomamba: Advancing dynamic facial expression recognition with visual and textual fusion. In *2025 IEEE International Conference on Image Processing (ICIP)*.
- Nguyen, L. P., Mille, J., Li, D. H., Conte, D., & Ragot, N. (2022). Efficient dynamic texture classification with probabilistic motifs. In *2022 26th International Conference on Pattern Recognition (ICPR)* (pp. 564–570). doi:10.1109/ICPR56361.2022.9956557
- Nguyen, L. P., Mille, J., Li, D., Conte, D., & Ragot, N. (2020). Trajectory extraction and deep features for classification of liquid-gas flow under the context of forced oscillation. In *15th International Conference on Computer Vision Theory and Applications* (pp. 17–26).
- Nguyen, L. P., Saleh, M., & Jeannès, R. L. B. (2017). An efficient design of a machine learning-based elderly fall detector. In *International Conference on IoT Technologies for Healthcare* (pp. 34–41). Springer.

SKILLS

Programming Languages

Python	C/C++	Matlab
Swift	LaTeX	Bash
Javascript	HTML/CSS	

Tools – Frameworks

OpenCV	Hugging Face	PyTorch
Git	Google Cloud	ARKit
vLLM	Keras	Linux

TEACHING EXPERIENCES

5th year École Polytechnique de l'Université de Tours

2020 – Neural networks – Deep Learning (Lectures and practical sessions)

3rd year INSA Centre Val de Loire - GSI

2019–2020 – Object oriented programming – C++ (Practical sessions)

Master 2 of Big Data Management and Analytics

2018–2019 – Introduction to Deep Learning (Practical sessions)

DISTINCTIONS

Scholarship from "Rencontre du Vietnam" organisation and Odon Vallet

📅 2016–2018

📍 France

National Olympiad in Physics

Silver medal

📅 2014

📍 Vietnam

EDUCATION

PhD in Computer Vision

University of Tours

📅 Nov 2018 – Dec 2021

Subject: **Extraction and Characterization of Spatial-Temporal Patterns in Videos**

Engineering Student

INSA Centre Val de Loire, campus Blois

📅 Sep 2015 – Sep 2018

Option: **Industrial Automatic Systems and Computer Science**

LANGUAGES

French Fluent – level B2

English Fluent – Advanced level

Vietnamese Mother tongue